Kehan Guo

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RESEARCH PROFILE

Ph.D. candidate at Notre Dame working on RL post-training and reward design for LLM agents. One question drives the work: when a model is optimized against a proxy — a reward model, a verifier, a frozen environment — where does the pressure actually go? Two answers are under review — a zero-cost reward interface in flow matching's noise–data coupling [5] and a live evaluation harness for tool-using agents [6] — both grown from community-standard benchmarks I created and co-led [1, 3]. Joining Apple as a research intern, June 2026; previously Amazon AWS AI and ByteDance.

EDUCATION

University of Notre Dame Aug 2022 – May 2027 (expected)
Ph.D. in Computer Science

- Advisor: Prof. Xiangliang Zhang (MINE Lab)

Boston University 2020 – 2022
M.S. in Computer Science

RESEARCH EXPERIENCE

Apple Jun 2026 – present
Research Intern · Cupertino, CA

- Research area: reinforcement-learning post-training for large language models.

ByteDance — Business Integrity Apr 2026 – Jun 2026
Research Scientist Intern · San Jose, CA

- Built BI-Claw, an internal tool-using LLM agent for business-integrity risk investigation, orchestrating 20+ domain skills over internal data services.
- Designed a self-evolving skill pipeline — skills seeded from static evaluation, evolved from real user queries with LLM-as-judge feedback — raising task success from 62% to 70% on the team's 145-case internal benchmark.
- Built the agentic evaluation harness behind that number — the industrial counterpart of [6].

Amazon AWS AI — Deep Engine Team May 2025 – Aug 2025
Applied Scientist Intern · New York, NY

- Designed **Social-Choice GRPO**, a reward shaping framework that converts sparse binary RL signals into dense rank-based feedback via social choice aggregation across policy rollouts, reducing gradient variance on long-horizon reasoning chains.
- Built a model-agnostic critique layer fusing rule-based verification with calibrated model confidence, enabling stable policy gradient updates on ambiguous, multi-step reasoning paths.
- Improved accuracy and sample efficiency over PPO baselines across math, coding, and logical reasoning tasks.

University of Notre Dame — MINE Lab Aug 2022 – Present
Research Assistant

Reward-Aligned Generation — Reward Transport [5] (under review)

- *Can alignment signal enter generative training through interfaces we already have?* Showed the noise–data coupling in flow matching — long treated as a computational convenience — is a zero-cost interface for property alignment, and characterized exactly when the interface exists.

Live Agentic Evaluation — ChemElucid-Live [6] (under review)

- *What do benchmark scores miss once the world moves?* Built a live evaluation harness measuring how tool-using agents acquire, verify, and act on information — the live complement to static benchmarks.

*Multimodal Reasoning — MolPuzzle (NeurIPS 2024 **[Spotlight]**) [1]*

- *Can frontier models do expert perception, not just expert recall?* Created MolPuzzle, a multimodal benchmark for molecular structure elucidation from spectra — now a community standard for probing the perception–reasoning gap.

LLM Evaluation for Science — ChemLLMBench (NeurIPS 2023) [3]

- *What can LLMs actually do in chemistry?* Co-led the first comprehensive benchmark of LLM chemistry capabilities (eight tasks) — now a standard reference for the field.

Selected Collaborations

- Co-authored TRACE (runtime preference enforcement for coding agents, 2026); contributed ideation and pipeline design to AdaReasoner (adaptive inference-time reasoning, NeurIPS 2025 Spotlight) and ChemOrch (74-tool agent orchestration, NeurIPS 2025).

SELECTED PUBLICATIONS

* equal contribution. Full list at Google Scholar.

[1] **K. Guo**, B. Nan, Y. Zhou, et al. "Can LLMs Solve Molecule Puzzles? A Multimodal Benchmark for Molecular Structure Elucidation." *NeurIPS 2024.* **[Spotlight]**

[2] Z. Song*, J. Lu*, Y. Du, B. Yu, T. M. Pruyn, Y. Huang, **K. Guo***, et al. "Evaluating Large Language Models in Scientific Discovery." *arXiv, 2025.*

[3] T. Guo*, **K. Guo***, B. Nan, et al. "What Can Large Language Models Do in Chemistry? A Comprehensive Benchmark on Eight Tasks." *NeurIPS 2023.*

[4] **K. Guo**, Y. Shen, et al. "Artificial Intelligence in Spectroscopy: Advancing Chemistry from Prediction to Generation and Beyond." *IJCAI 2025.* **[Survey Track]**

Preprints / Under Review

[5] **K. Guo**, Y. Shen, Y. Zhou, Y. Huang, C. Gao, S. Du, X. Zhang. Reward Transport: Property Control in Flow Matching via Noise-Space Alignment. *Under review.* [Code]

[6] **K. Guo**, et al. ChemElucid-Live — live evaluation of tool-using scientific agents. *Under review.*

TECHNICAL SKILLS

Languages	Python, C++, SQL, Bash
Training	PyTorch, JAX, Megatron-LM, DeepSpeed ZeRO, Ray, VeRL, GRPO/PPO/DPO pipelines, reward modeling
Inference & Agents	vLLM, HF Transformers, function-calling, Neo4j
Dev Tools	AI-assisted development: Claude Code, Codex
Scientific	RDKit, PubChem, spectroscopy processing

AWARDS & RECOGNITION

NeurIPS 2024 Spotlight — *MolPuzzle* (first author)

NeurIPS 2025 Spotlight — *AdaReasoner* (co-author)

OpenAI Researcher Access Program (2024)

PROFESSIONAL SERVICE

Reviewer: NeurIPS (2024–25), ICLR (2025–26), ICML, AAAI, IJCAI, KDD, ACL Rolling Review, WWW, EMNLP